

Constraints: Ensure that the total thickness of the layers does not exceed 7 millimeters. If the constraint is violated, apply a penalty to the cost function.

Objective: The primary goal is to maximize the radar frequency reflection. This will be represented by minimizing the negative reflection coefficient.

Combine Constraints and Objectives into the cost function: Create cost function that penalizes any solutions that exceed the thickness limit while rewarding solutions that achieve high reflection coefficients.

Cost function: minimize thickness while maximizing absorption

Check thickness constraint

```
total_thickness = jnp.sum(d)
```

Penalty for exceeding the thickness limit

```
thickness_penalty = jnp.where(total_thickness > thickness_limit, 1e6 * (total_thickness -  
thickness_limit)**2, 0.0)
```

^,0.0 is Boolean condition in the jnp.where function

Adjust the penalty factor (1e6) based on how severely to penalize violations of the thickness constraint.

Non-linearly squaring the penalty term introduces non-linearity, which can help the optimization algorithm differentiate between solutions that slightly exceed the constraint and those that greatly exceed it. This helps to prioritize more severe violations.

Question: do we square the penalty?

Ex:

```
def cost_function(n, d, f, thickness_limit=7e-3):
```

Variables to have in cost function – reflection and absorption

Calculate the reflection coefficients

```
R_TE, T_TE, R_TM, T_TM = stackrt(n, d, f)
```

R_TE, T_TE, R_TM, T_TM represent the reflection and transmission coefficients for two types of polarization

Function that calculates the reflection and transmission coefficients?

R_TE at a specific radar frequency index

```
radar_frequency_index = 0
```

```
reflection_coefficient = R_TE[radar_frequency_index]
```

^^line above is used to extract the reflection coefficient for a specific radar frequency from the array of reflection coefficients.

The cost is the negative reflection coefficient to maximize it, plus the added thickness penalty

cost = -reflection_coefficient + thickness_penalty

Calculate the cost for the optimization algorithm

Parameters:

- n: Refractive indices for the layers.
- d: Thickness of the layers.
- f: Frequencies at which to calculate the coefficients.
- thickness_limit: Maximum allowed thickness in meters (7 mm). Returns:

- cost function output: lower is better.